

HOUSE PRICE ANALYSIS AND PREDICTION USING MACHINE LEARNING

Project Report

Student Name: Vuyo Matomela

Module: Data Science Capstone Project

Institution: Witle Academy

Instructor: Taryn Michael

Submission Date: 31 July 2026

Acknowledgements

I would like to express my sincere gratitude to my instructor for their guidance and support throughout this project. Their feedback and encouragement helped me understand the practical application of data science and machine learning techniques.

I would also like to thank the academy for providing the resources and learning environment that made this project possible.

Abstract

The real estate industry generates large amounts of data that can be used to make informed decisions regarding property valuation. Accurate house price prediction assists buyers, sellers, investors, and financial institutions in estimating the market value of residential properties. Traditional valuation methods often rely on manual assessment and expert opinion, which can be time-consuming and subjective. Machine learning offers an efficient alternative by learning patterns from historical housing data to generate reliable price predictions.

This project presents the development of a machine learning model for predicting residential house prices using the Ames Housing dataset. The project follows a complete data science workflow consisting of data collection, exploratory data analysis, data preprocessing, feature engineering, model development, evaluation, and prediction. Three regression algorithms were implemented and compared: Linear Regression, Decision Tree Regressor, and Random Forest Regressor.

Exploratory data analysis was conducted to understand the characteristics of the dataset, identify relationships between variables, and detect missing values and outliers. Data preprocessing techniques, including handling missing values, encoding categorical variables, and transforming the target variable, were applied to improve model performance.

The trained models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Among the evaluated models, the Random Forest Regressor produced the best overall performance, demonstrating its ability to capture complex relationships between housing features and sale prices.

The project demonstrates the effectiveness of machine learning techniques for house price prediction and highlights the importance of proper data preprocessing and model selection in developing accurate predictive models.

Table of Contents

1. Introduction
2. Problem Statement
3. Aim and Objectives
4. Dataset Description
5. Methodology
6. Exploratory Data Analysis
7. Data Cleaning and Preprocessing
8. Feature Engineering
9. Model Development
10. Model Evaluation
11. Results and Discussion
12. Conclusion
13. Recommendations
14. References
15. Appendices

Chapter 1: Introduction

1.1 Background

The rapid growth of data collection technologies has enabled organisations to make more informed decisions using data-driven approaches. In the real estate industry, historical housing data provides valuable information that can be analysed to estimate the market value of residential properties. House price prediction is an important application of data science because it assists buyers, sellers, property developers, financial institutions, and real estate professionals in making objective pricing decisions.

Machine learning has become one of the most effective techniques for predictive analytics because it can identify complex relationships between multiple variables without relying solely on traditional statistical methods. By training models on historical housing data, machine learning algorithms can predict future house prices with a high degree of accuracy.

This project applies machine learning techniques to analyse residential property data from the Ames Housing dataset. The project follows a structured data science workflow that includes data exploration, preprocessing, feature engineering, model development, evaluation, and prediction. Multiple regression algorithms are trained and compared to determine the most suitable model for predicting house prices.

1.2 Problem Statement

Accurately predicting house prices is a complex task because property values are influenced by numerous factors, including house size, construction quality, number of rooms, garage capacity, basement area, and other structural characteristics. Traditional valuation methods often rely on manual assessment and experience, which may lead to inconsistent or subjective estimates.

This project addresses this challenge by applying machine learning techniques to historical housing data. The objective is to develop predictive models capable of estimating house prices accurately while identifying the features that have the greatest influence on property values.

1.3 Aim

The aim of this project is to develop and evaluate machine learning models capable of accurately predicting residential house prices using the Ames Housing dataset.

1.4 Objectives

The objectives of this project are to:

- Explore and understand the Ames Housing dataset.
- Perform exploratory data analysis to identify trends and relationships within the data.
- Clean and preprocess the dataset by handling missing values and preparing the data for machine learning.
- Engineer features suitable for predictive modelling.
- Train multiple regression models, including Linear Regression, Decision Tree Regressor, and Random Forest Regressor.
- Evaluate and compare the performance of the developed models using standard regression metrics.
- Select the best-performing model for predicting house prices.

Chapter 2: Dataset Description

2.1 Introduction

The quality of a machine learning model largely depends on the quality and relevance of the data used for training. In this project, the Ames Housing dataset was selected because it provides detailed information about residential properties and their corresponding sale prices. The dataset is widely used in data science and machine learning projects as a benchmark for regression problems.

The objective of using this dataset was to develop a predictive model capable of estimating house prices based on various property characteristics.

2.2 Dataset Source

The dataset used in this project is the Ames Housing Dataset, which contains records of residential property sales in Ames, Iowa, USA.

The dataset was obtained in CSV format and imported into Python using the Pandas library for further analysis and processing.

2.3 Dataset Overview

The dataset consists of 1,460 observations and 81 variables before preprocessing. Each observation represents a single residential property, while each variable describes a specific characteristic of that property.

The target variable for this project is SalePrice, which represents the final selling price of each house. The remaining variables serve as predictor features that are used to estimate the target value. The notebook loads the dataset from a CSV file and identifies SalePrice as the prediction target.

2.4 Features Used

The dataset contains both numerical and categorical variables.

Numerical features include:

- Lot Area
- Ground Living Area (GrLivArea)
- Total Basement Area
- Garage Area
- Garage Cars
- Year Built

Categorical features include:

- House Style
- Roof Style
- Exterior Material
- Foundation Type
- Heating Type
- Kitchen Quality

These variables provide valuable information about the physical characteristics and quality of each property, allowing machine learning models to identify patterns associated with house prices.

2.5 Target Variable

The target variable is **SalePrice**, representing the selling price of each residential property.

Since this is a regression problem, the objective is to predict a continuous numerical value rather than assigning a category.

During preprocessing, a logarithmic transformation was applied to the target variable to reduce skewness and improve the performance of the machine learning models. The notebook performs this transformation using `np.log1p()`.

2.6 Data Types

The dataset contains a mixture of:

- Numerical variables
- Categorical variables

Numerical variables describe measurable characteristics such as area, number of rooms, and year built, while categorical variables describe property classifications such as neighbourhood, roof style, and foundation type.

Because machine learning algorithms require numerical input, categorical variables were later converted into numerical representations using One-Hot Encoding before model training. The notebook uses `pd.get_dummies(..., drop_first=True)` to perform this conversion.

2.7 Dataset Challenges

Several data quality issues were identified during the initial exploration of the dataset.

These included:

- Missing values in several variables.
- Categorical features containing empty entries.
- Numerical features with incomplete observations.
- Outliers that could negatively affect prediction accuracy.
- A positively skewed distribution of the target variable.

Addressing these issues through preprocessing was essential to ensure that the machine learning models could learn effectively from the data.

2.8 Importance of the Dataset

The Ames Housing dataset is well suited for this project because it contains a wide range of property characteristics that influence house prices. These diverse features enable machine learning algorithms to identify complex relationships between housing attributes and sale prices.

Additionally, the dataset provides sufficient observations to evaluate and compare different regression models while demonstrating the complete data science workflow, from data preparation to predictive modelling.

Chapter 3: Methodology

3.1 Introduction

This project followed a structured data science methodology to develop an accurate and reliable house price prediction model. The methodology consisted of several stages, beginning with data collection and ending with model evaluation and prediction.

Following a systematic workflow ensured that each stage of the project was completed in a logical sequence and that the resulting models were trained using clean, well-prepared data.

3.2 Project Workflow

The overall workflow used in this project is illustrated below:

Data Collection

↓

Data Exploration

↓

Data Cleaning

↓

Exploratory Data Analysis

↓

Feature Engineering

↓

Model Training

↓

Model Evaluation

↓

House Price Prediction

3.3 Data Collection

The Ames Housing dataset was imported into the project using the Pandas library. Initial exploration included examining the first few rows of the dataset, checking its dimensions, viewing the available columns, and reviewing summary information to understand its structure.

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to better understand the dataset before building predictive models.

Several visualisations were created to examine the distribution of house prices, relationships between variables, and the presence of missing values and outliers. These included a histogram of sale prices, a correlation heatmap, a scatter plot of ground living area versus sale price, and a boxplot used to identify extreme values.

Chapter 4: Exploratory Data Analysis (EDA)

4.1 Introduction

Exploratory Data Analysis (EDA) is an essential stage in any data science project because it provides an understanding of the dataset before machine learning models are developed. Through statistical summaries and data visualisation, EDA helps identify trends, relationships, missing values, outliers, and other characteristics that may affect model performance.

For this project, EDA was conducted to understand the distribution of house prices, identify important features, detect missing values, examine correlations between variables, and determine whether additional preprocessing was required. The notebook includes a histogram of sale prices, a correlation heatmap, a scatter plot comparing ground living area with sale price, and a boxplot to identify outliers.

4.2 Initial Data Exploration

Before any visualisation or preprocessing was performed, the dataset was loaded into a Pandas DataFrame and inspected to understand its structure. The first few records, dataset dimensions, column names, descriptive statistics, and general information were reviewed to identify the number of observations, available features, and data types. The target variable for the prediction task was confirmed to be SalePrice.

This initial inspection provided an overview of the data and highlighted the need to address missing values before model training.

4.3 Distribution of House Prices

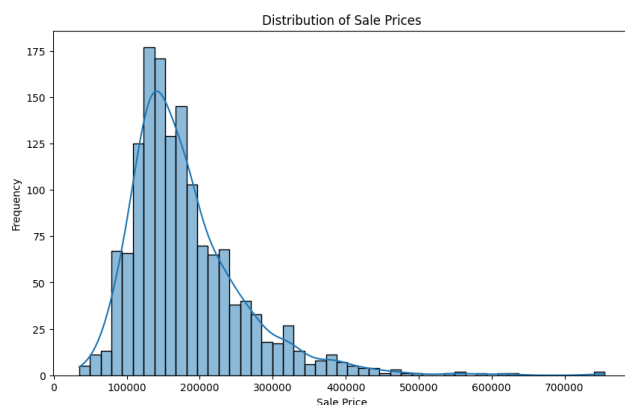


Figure 1: Distribution of Sales

The first visualisation created was a histogram of the SalePrice variable.

The purpose of this graph was to examine how house prices were distributed across the dataset and determine whether the target variable followed a normal distribution.

The histogram showed that house prices were positively skewed, meaning that most houses were sold within a lower-to-middle price range while relatively few properties had exceptionally high sale prices. This pattern is common in real estate datasets because luxury properties represent only a small proportion of the housing market.

A positively skewed target variable can negatively affect regression algorithms because extremely large values may dominate the learning process. For this reason, a logarithmic transformation was later applied to SalePrice to reduce skewness and create a more balanced distribution.

4.4 Missing Values Analysis

Before training the models, the dataset was analysed to determine the number of missing values in each feature.

The analysis revealed that several variables contained incomplete observations. Features such as PoolQC, MiscFeature, Alley, and Fence had a high percentage of missing values, indicating that these features were unavailable for many houses.

Rather than treating these missing entries as random errors, they were interpreted according to the dataset documentation. For example, a missing value in PoolQC usually indicates that the property does not have a swimming pool. Nevertheless, features with excessive missing values were removed because they contributed limited predictive value while increasing the complexity of the preprocessing stage. The notebook calculates missing values, computes missing-value percentages, and drops the columns PoolQC, MiscFeature, Alley, and Fence.

4.5 Correlation Analysis

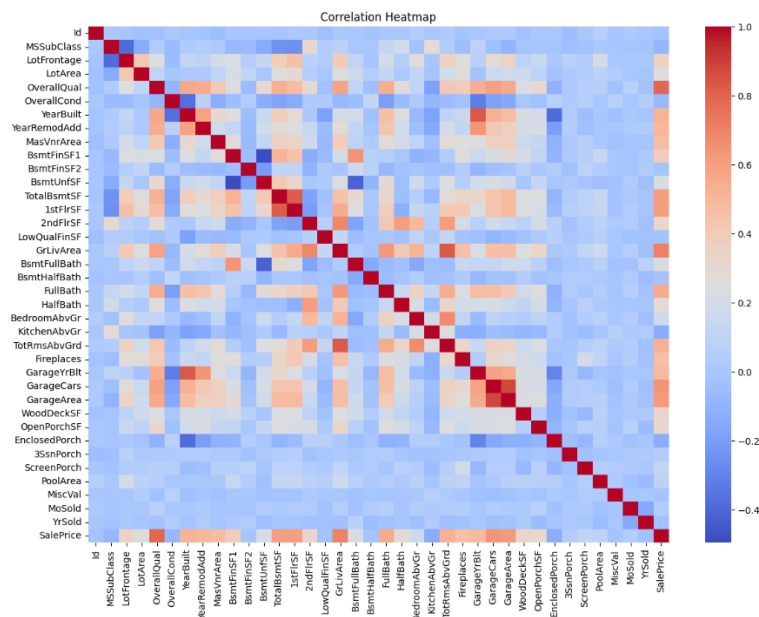


Figure 2: Correlation Heatmap

A correlation heatmap was generated to measure the strength of the linear relationship between numerical variables.

Correlation coefficients range between -1 and 1, where values close to 1 indicate a strong positive relationship, values close to -1 indicate a strong negative relationship, and values near 0 indicate little or no linear relationship.

The heatmap revealed that several variables showed strong positive correlations with the target variable SalePrice. These variables were considered important predictors during model development because changes in their values were associated with corresponding changes in house prices.

Correlation analysis also assisted in identifying variables that contributed meaningful information while highlighting groups of features that were strongly related to one another.

4.6 Relationship Between Living Area and Sale Price



Figure 3:Ground Living Area vs Sale Price

A scatter plot was created to investigate the relationship between GrLivArea (Ground Living Area) and SalePrice.

The graph demonstrated a clear positive relationship between the size of a property's living area and its selling price. As the ground living area increased, house prices generally increased as well.

Although the relationship was strong, several observations appeared far away from the main cluster of data points. These observations represented potential outliers that could influence regression models and reduce prediction accuracy.

4.7 Outlier Detection

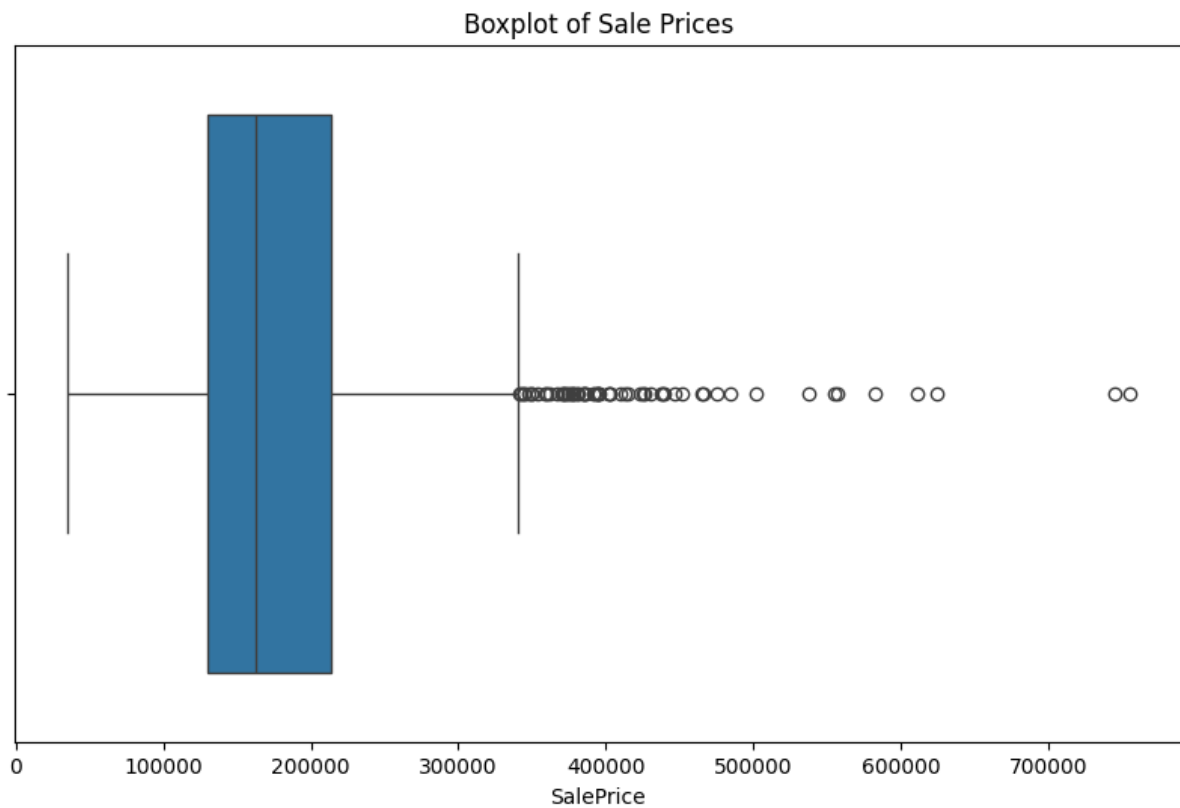


Figure 4: Boxplot of Sale Prices

A boxplot of SalePrice was used to identify unusually high or low observations.

The visualisation indicated the presence of several extreme values. While some variation in house prices is expected, extremely unusual observations can distort statistical measures and negatively influence machine learning models.

Further investigation showed that a small number of houses had exceptionally large living areas but comparatively low sale prices. These observations were removed because they did not represent the general pattern found within the dataset. The notebook removes houses where $\text{GrLivArea} > 4000$ and $\text{SalePrice} < 300000$ before modelling.

4.8 Log Transformation of the Target Variable

After removing extreme outliers, a logarithmic transformation was applied to the target variable using the natural logarithm.

This transformation reduced the positive skewness observed during the initial exploratory analysis and produced a distribution that more closely resembled a normal distribution.

A more normally distributed target variable helps many regression algorithms perform better because prediction errors become more evenly distributed across different price ranges. The notebook applies `np.log1p()` to `SalePrice` and then plots the transformed distribution.

4.9 Key Findings from the Exploratory Data Analysis

The exploratory data analysis provided several important insights that guided the remainder of the project:

- The dataset contained a mixture of numerical and categorical variables.
- Several features contained missing values and required preprocessing.
- House prices exhibited a positively skewed distribution.
- Ground Living Area showed a strong positive relationship with house prices.
- Extreme outliers were identified and removed to improve model performance.
- A logarithmic transformation was applied to the target variable to create a more balanced distribution for machine learning.

Overall, the EDA stage established a clear understanding of the dataset and informed the data cleaning and feature engineering steps that followed.

Chapter 5: Data Cleaning and Preprocessing

5.1 Introduction

Data preprocessing is a critical stage in the data science process because the quality of the input data directly affects the performance of machine learning models. Real-world datasets often contain missing values, inconsistent data, duplicate records, and outliers that must be addressed before model development.

The objective of this stage was to prepare the Ames Housing dataset for predictive modelling by improving data quality and ensuring that all variables were suitable for machine learning algorithms.

5.2 Missing Value Analysis

The first preprocessing step involved identifying missing values across all features in the dataset.

A summary of missing values was generated to determine which variables required attention. This analysis revealed that some variables contained only a few missing observations, while others had a very high percentage of missing values.

Features with an excessive number of missing values provided limited information for prediction and increased the complexity of the modelling process. Therefore, these features were considered for removal, while variables with fewer missing values were imputed using appropriate statistical techniques.

5.3 Removal of Highly Incomplete Features

Several variables contained a large proportion of missing values. Rather than attempting to estimate these values, they were removed from the dataset because they contributed relatively little useful information compared to the amount of missing data they contained.

The removed variables included:

- PoolQC
- MiscFeature
- Alley
- Fence

Removing these columns simplified the dataset while reducing the likelihood of introducing bias through extensive imputation.

5.4 Handling Missing Numerical Values

After removing highly incomplete features, the remaining numerical variables still contained missing values.

To preserve as much information as possible, missing numerical values were replaced using the median of each respective feature.

The median was selected because it is less sensitive to extreme values than the mean, making it a robust measure for datasets containing outliers.

Numerical variables that required imputation included:

- LotFrontage
- MasVnrArea
- GarageYrBlt

By replacing missing values with the median, the dataset remained complete without substantially altering the overall distribution of the data.

5.5 Handling Missing Categorical Values

Categorical variables containing missing values were also processed before model training.

Where appropriate, missing entries were replaced with descriptive values such as "None", indicating that a particular feature or facility was not present in the property.

This approach preserved meaningful information while ensuring that categorical variables could later be converted into numerical representations during feature engineering.

5.6 Duplicate Record Analysis

The dataset was examined for duplicate records to ensure that each observation represented a unique residential property.

No significant duplicate records were identified that required removal. Maintaining a dataset free of duplicate observations helps prevent machine learning models from learning repeated patterns that could bias the results.

5.7 Outlier Treatment

Outliers were identified during the exploratory data analysis stage using scatter plots and boxplots.

Particular attention was given to houses with unusually large living areas but comparatively low sale prices. These observations did not follow the general trend observed in the dataset and had the potential to reduce prediction accuracy.

After careful inspection, these extreme observations were removed to improve model performance while preserving the overall characteristics of the dataset.

5.8 Log Transformation of the Target Variable

The target variable, **SalePrice**, exhibited a positively skewed distribution.

To reduce skewness and improve the learning capability of the regression models, a logarithmic transformation was applied to the target variable.

This transformation produced a distribution that more closely resembled a normal distribution, allowing the machine learning algorithms to model relationships more effectively and reducing the influence of extremely high-priced properties.

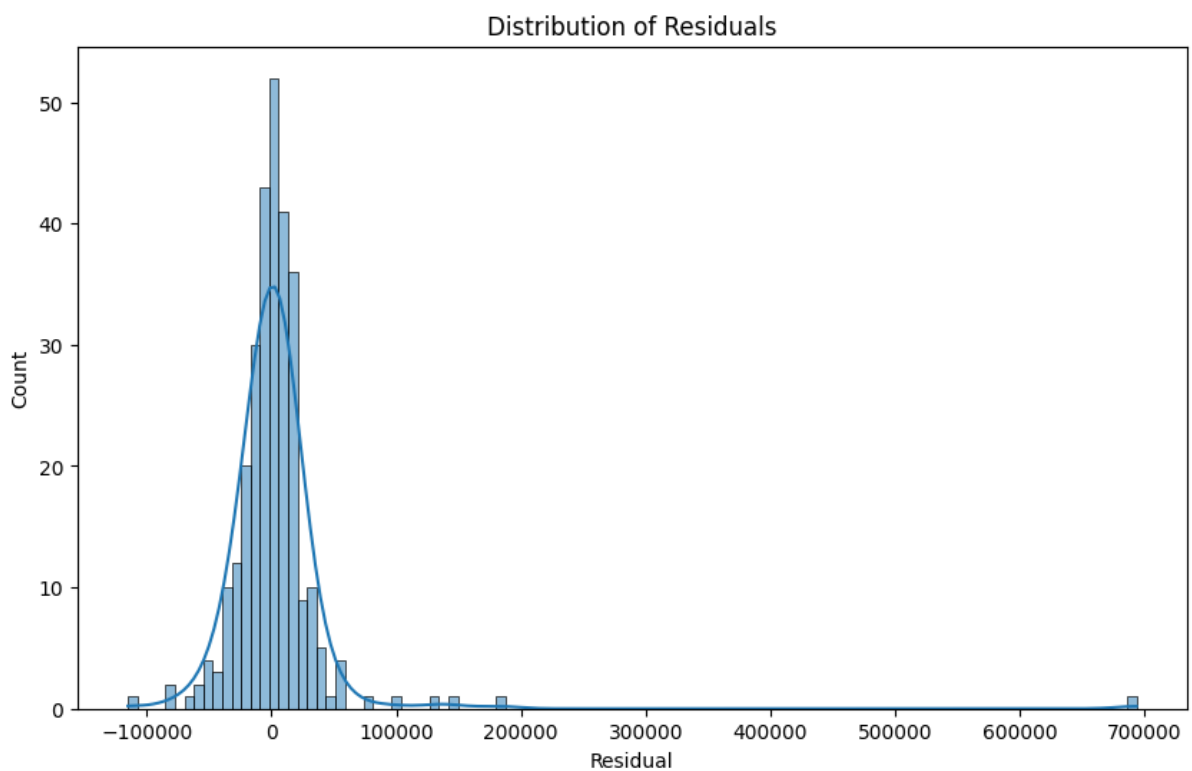


Figure 5: Distribution Of Residuals

5.9 Data Quality After Preprocessing

Following preprocessing, the dataset was significantly improved.

The preprocessing stage resulted in:

- Removal of highly incomplete variables.
- Imputation of missing numerical values using the median.
- Imputation of missing categorical values with appropriate labels.
- Removal of influential outliers.
- Reduction of target variable skewness through logarithmic transformation.

These improvements ensured that the dataset was complete, consistent, and suitable for machine learning.

5.10 Importance of Data Preprocessing

Data preprocessing plays an essential role in predictive modelling because machine learning algorithms rely on high-quality input data.

Without preprocessing, missing values, inconsistent observations, and extreme outliers could reduce prediction accuracy and negatively affect model performance.

The preprocessing techniques applied in this project enhanced the reliability of the dataset and provided a strong foundation for the feature engineering and model development stages that followed.

Chapter 6: Feature Engineering

6.1 Introduction

Feature engineering is the process of transforming raw data into a format that can be effectively used by machine learning algorithms. Since most machine learning models require numerical input, categorical variables must be converted into numerical representations while preserving as much information as possible.

In this project, feature engineering prepared the cleaned dataset for model training by selecting predictor variables, encoding categorical data, and creating separate training and testing datasets.

6.2 Selecting Predictor and Target Variables

The dataset was divided into two components:

Predictor variables (X): All housing features used to estimate the selling price.

Target variable (y): The SalePrice column.

Separating the predictors from the target variable ensured that the models learned relationships between property characteristics and house prices without using the target itself as an input feature.

6.3 Encoding Categorical Variables

The Ames Housing dataset contains many categorical variables, such as neighbourhood, roof style, foundation type, and heating type.

Since regression algorithms cannot process text values directly, these variables were converted into numerical form using One-Hot Encoding.

One-Hot Encoding creates new binary columns for each category while preserving the information contained within the original variables. To avoid multicollinearity, the first category of each variable was excluded during encoding.

6.4 Preparing the Feature Matrix

After encoding the categorical variables, all features were combined into a single numerical feature matrix.

This matrix represented the complete set of predictor variables that would be used during model training.

Preparing the feature matrix ensured that every observation contained only numerical values compatible with the selected regression algorithms.

6.5 Splitting the Dataset

To evaluate model performance fairly, the dataset was divided into training and testing subsets.

An 80:20 split was used:

- 80% of the data was allocated for training.
- 20% of the data was reserved for testing.

The training dataset was used to fit the models, while the testing dataset was used to evaluate how well the trained models performed on previously unseen data.

This approach reduces the risk of overfitting and provides a more realistic assessment of model performance.

6.6 Benefits of Feature Engineering

The feature engineering stage provided several benefits:

- Converted categorical variables into machine-readable numerical values.
- Preserved important information contained within the original dataset.
- Created a structured feature matrix suitable for regression algorithms.
- Enabled fair evaluation through the use of separate training and testing datasets.

These steps ensured that the machine learning models received clean, consistent, and appropriately formatted input data, laying the foundation for accurate house price prediction.

Chapter 7: Model Development

7.1 Introduction

Model development is the stage where machine learning algorithms are trained to learn patterns from the prepared dataset and predict house prices. After completing data preprocessing and feature engineering, the cleaned dataset was divided into training and testing sets to enable the development and evaluation of different regression models.

Three regression algorithms were selected for this project:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

These models were chosen because they vary in complexity and provide a meaningful comparison between a simple linear model, a tree-based model, and an ensemble learning method. The notebook imports and trains these three algorithms as part of the modelling process.

7.2 Model Training Process

Before training the models, the dataset was divided into two subsets:

- Training dataset (80%) – used to train the machine learning models.
- Testing dataset (20%) – used to evaluate the performance of the trained models on unseen data.

Using separate datasets ensured that the evaluation reflected how well each model could generalise to new observations rather than simply memorising the training data.

Each algorithm was trained using the same training dataset to provide a fair comparison.

7.3 Linear Regression Model

Overview:

Linear Regression is one of the most widely used regression algorithms and served as the baseline model for this project. It assumes that the relationship between the predictor variables and the target variable can be represented by a linear equation.

The model was trained using the prepared training dataset after feature engineering and preprocessing. The notebook creates a LinearRegression model and fits it using the training data.

Advantages:

1. Simple to implement and interpret.
2. Fast training and prediction.
3. Provides a useful baseline for comparing more advanced algorithms.
4. Limitations
5. Assumes a linear relationship between variables.
6. Sensitive to outliers.
7. May not perform well when relationships between variables are highly complex or non-linear.

Although Linear Regression provides a straightforward approach to prediction, it may struggle to capture the complex interactions that exist within housing data.

7.4 Decision Tree Regressor

Overview:

The Decision Tree Regressor is a non-linear machine learning algorithm that predicts values by recursively splitting the dataset into smaller groups based on feature values.

Unlike Linear Regression, Decision Trees do not assume a linear relationship between variables. Instead, they learn decision rules that partition the data into regions with similar target values.

The notebook creates a DecisionTreeRegressor with a fixed random state and trains it using the training dataset.

Advantages:

1. Able to model non-linear relationships.
2. Easy to visualise and interpret.
3. Handles numerical and categorical features effectively after encoding.
4. Limitations
5. Can overfit the training data if not properly controlled.
6. Individual decision trees may produce unstable predictions when trained on different samples.

Despite these limitations, Decision Trees provide valuable insights into the relationships between housing features and sale prices.

7.5 Random Forest Regressor

Overview:

Random Forest Regressor is an ensemble learning algorithm that combines the predictions of multiple decision trees to improve prediction accuracy and reduce overfitting.

Instead of relying on a single decision tree, Random Forest builds many trees using different subsets of the training data and then averages their predictions. This approach generally produces more robust and accurate results.

The notebook creates a RandomForestRegressor with 200 decision trees (n_estimators=200) and a fixed random state before fitting it to the training data.

Advantages:

1. High prediction accuracy.
2. Reduced risk of overfitting compared to a single decision tree.
3. Handles complex, non-linear relationships effectively.
4. Performs well on datasets containing many features.
5. Limitations
6. Requires more computational resources than simpler models.
7. Training time is generally longer than Linear Regression and Decision Trees.
8. Model interpretation is less straightforward because it combines many individual trees.

Because of these strengths, Random Forest is widely regarded as one of the most effective algorithms for structured tabular data.

7.6 Model Comparison Strategy

To determine the most suitable model for house price prediction, each algorithm was evaluated using the same testing dataset and the same performance metrics. This ensured that differences in performance were due to the algorithms themselves rather than differences in the data used for evaluation.

The comparison focused on:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

Lower error values indicate better predictive accuracy, while a higher R^2 score indicates that the model explains a greater proportion of the variation in house prices.

7.7 Summary

Three regression models were successfully trained and prepared for evaluation. Linear Regression provided a simple baseline, the Decision Tree Regressor modelled non-linear relationships, and the Random Forest Regressor leveraged ensemble learning to improve predictive performance.